

ANTHONY JARJOUR B.S.

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EDUCATION	California State University, Fullerton California, United States <i>B.S. Computer Science, Minor in Mathematics and Physics</i> 2021.08 - 2026.05 Extended degree by one year to formally pursue advanced physics and mathematics coursework and research in theoretical physics.
RESEARCH	Undergraduate Research Assistant Theoretical Physics – QFT 2025.05 – Present - Investigated thermodynamic properties of inequivalent Rindler vacua in Minkowski spacetime following Lochan et al. (2025), focusing on Bogoliubov transformations, vacuum inequivalence, and Planck-scale aspects of quantum field theory. - Built graduate-level theoretical foundation in General Relativity and Quantum Field Theory using Schutz (1985), Wald (1994), Ashtekar (1975), and Panangaden (1979), covering algebraic QFT, Green's functions and propagators, and Kähler structures.
PRESENTATION	Bogoliubov Transformations Between Shifted Rindler Frames 2026.01 <i>Joint Mathematics Meetings (JMM)</i> - Presented at the 2026 Joint Mathematics Meetings an expanded study of Bogoliubov transformations between shifted Rindler frames, exploring how different quantization schemes yield distinct vacuum structures and physical interpretations. Bogoliubov Transformations Between Shifted Rindler Frames 2025.10 <i>Math Research Symposium</i> - Delivered an in-depth exposition of Lochan & Padmanabhan (2025), reproducing their derivation of Bogoliubov transformations between shifted Rindler frames and analyzing the resulting inequivalent vacuum structure in detail. Data Science in Sport Analytics ECS Innovation Expo 2025.04 - Cleaned and preprocessed soccer match data to develop machine learning models for predicting game outcomes, and presented application at an expo.
AWARDS & HONORS	Dean's List California State University, Fullerton 2021.08 - 2026.05 - Recognized for academic excellence for the Fall 2024, Spring 2025, Fall 2025, and Spring 2026 semesters Funded MSRP Award CSUF Department of Mathematics 2025.05 - Selected for the CSUF Department of Mathematics Math Summer Research Program to conduct 100 hours of paid undergraduate research. - Worked under faculty mentorship on a mathematics research project, culminating in a research poster and summary of findings. 3rd Place Award CSUF CS Showcase 2025.04 - Awarded 3rd place for developing a machine learning model to predict soccer match outcomes, presented at the Computer Science Showcase at California State University, Fullerton.

TEACHING EXPERIENCE	Mathematical Circle Instructor - CSUF Fullerton, California 2026.01 - 2026.05	
	• Lead a problem-solving session for middle- and high-school students in an open university outreach program, delivering a lecture and workshop-style activities. • Designed and curated challenging problem sets and guided solutions to strengthen mathematical reasoning, proof skills, and creativity. • Collaborated with faculty mentorship to select topics, refine lesson plans, and adapt instruction to varied student backgrounds and ability levels.	
	Undergraduate Tutor (Volunteer) - CSUF Fullerton, California 2025.08 - 2025.12	
	• Mentored undergraduate students through homework and exam preparation in vector and tensor analysis. • Broke down complex derivations into approachable steps and emphasized geometric/physical interpretation alongside formalism. • Developed targeted practice problems and review notes to address recurring conceptual gaps.	
RELEVANT COURSEWORK	Completed	
	Computer Science: Compilers and Languages; Algorithm Engineering; Theory of Computation; Artificial Intelligence; Machine Learning Mathematics: Vector and Tensor Analysis; Mathematical Probability; Partial Differential Equations; Differential Geometry; Quantum Field Theory; Intro to General Relativity Physics: Thermodynamics, Kinetic Theory, and Statistical Physics; Classical Dynamics	
	Planned	
	Real Analysis I–II; Linear Algebra; Complex Analysis; Electromagnetic Theory I–II; Quantum Mechanics I–II	
ACADEMIC INTERESTS	Theoretical Physics, Quantum Field Theory, General Relativity, Quantum Gravity, Computational Physics, Machine Learning for Physical Systems.	
PROJECTS	Large Language Model for Code Generation 2024.09 – 2024.11	
	• Technologies: Python, NLTK, Numpy, PyTorch, Pandas • Notes: Evaluated model output using unit tests to validate code correctness across multiple functional tasks, with a 93% pass rate.	
	N-Body Problem Simulator 2024.08 – 2024.10	
	• Technologies: Python, Numpy, Matplotlib • Notes: Implemented Runge-Kutta integration to model two-body dynamics in a Newtonian gravitational system; visualized orbit evolution and field behavior. • Simulated gravitational fields using Newtonian mechanics with relativistic corrections	
	Residual Neural Network Image Classifier 2024.01 – 2024.05	
	• Technologies: Python, PyTorch, NodeJS, Javascript, FastAPI, ScikitLearn, Pandas • Notes: Implemented an RNN based on K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," arXiv preprint arXiv:1512.03385, 2015, and achieved 95% validation accuracy with 0.25 loss, on bird species images	

SKILLS

Programming Languages:

Python, C, C++, JavaScript, PHP, HTML, MATLAB, R

Machine Learning / Data Science:

PyTorch, Torchvision, NumPy, Pandas, Scikit-learn, Matplotlib, Hugging Face Hub, Jupyter Notebooks

Web / App Development:

React Native, Expo, Node.js, npm, FastAPI, Uvicorn, REST APIs

Tools / Systems:

Git, GitHub, Linux/Ubuntu, VS Code, SQLite/MySQL-style database design

INTERNSHIPS

Cybersecurity Analyst - El Paso Water Utility | El Paso, Texas 2022.06 - 2022.08

- Designed and automated reports to visualize trends across multidimensional data streams, enhancing system performance and predictive diagnostics.

SELECTED REFERENCES

- K. Lochan and T. Padmanabhan, "A Nested Sequence of Inequivalent Rindler Vacua: Universal Relic Thermalty of Planckian Origin," *Classical and Quantum Gravity*, vol. 42, no. 3, 03LT01, 2025.
- B. F. Schutz, *A First Course in General Relativity*. Cambridge University Press, 1985.
- R. M. Wald, *Quantum Field Theory in Curved Spacetime and Black Hole Thermodynamics*. University of Chicago Press, 1994.
- A. Ashtekar and A. Magnon, "Quantum Fields in Curved Space-Times," *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences*, vol. 346, no. 1646, pp. 375–394, 1975.
- P. Panangaden, "Positive and Negative Frequency Decompositions in Curved Spacetime," *Journal of Mathematical Physics*, vol. 20, no. 12, pp. 2506–2510, 1979.